

Q40 Capstone Project Summary

Observation

A production-style Artificial Neural Network (ANN) pipeline was developed for customer purchase prediction using the features Age, Salary, and Experience. The workflow included data validation, preprocessing using StandardScaler within a preprocessing pipeline, feature engineering, and a stratified train/validation/test split to preserve the class distribution. The ANN consisted of three hidden layers enhanced with Batch Normalization and Dropout to improve stability and reduce overfitting. Training incorporated EarlyStopping, ModelCheckpoint, and ReduceLROnPlateau callbacks to improve generalization and automatically retain the best-performing model. Model evaluation included a confusion matrix, classification report, ROC-AUC, Precision-Recall analysis, SHAP-based explainability, and accuracy/loss curves. The final trained model was saved and reloaded successfully, and a prediction function was implemented for new customer records.

Insight Interpretation

The experiments showed that proper preprocessing and feature scaling significantly improved optimization and convergence. Regularization techniques such as Dropout and Batch Normalization reduced overfitting, resulting in closer training and validation performance. EarlyStopping prevented unnecessary training epochs, while learning-rate scheduling improved convergence. Cross-validation demonstrated consistent performance across different folds, indicating good generalization. SHAP explanations revealed that Salary and Experience had the strongest influence on purchase probability, with Age contributing additional predictive information. Precision-Recall analysis proved especially useful because the dataset contained an intentional class imbalance, making it more informative than accuracy alone.

Business Recommendation

The proposed ANN pipeline can be deployed as a customer purchase prediction system within a CRM or marketing platform. Customers predicted to have a high purchase probability can receive personalized promotions, premium recommendations, or loyalty offers, while customers with lower probability can be targeted through retention campaigns and discounts. The saved TensorFlow model and prediction function enable straightforward integration into web or mobile applications. Regular retraining with fresh customer data and continuous monitoring of model performance are recommended to maintain accuracy as customer behavior changes. Explainability tools such as SHAP should be retained in production to improve transparency, support business decisions, and build stakeholder trust in the model's predictions.